The negative map:
seventeen pre-registered closures on the Möbius dilate field

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Abstract

Any attempt to *prove* the Möbius-twisted Elliott–Halberstam hypothesis EH_μ by dispersion arrives at the dilate field

$$D(k) \;=\; \sum_m \mu(m)\,\mu(N-mk),$$

in which the two Möbius arguments are coupled by a product and a difference simultaneously. This report records what happens when seventeen distinct routes into that field are specified in advance, given a decision rule and a null, and run: five adjudications of existing machinery, nine technique designs kill-tested numerically, and four representation classes. Sixteen close; one is reopened here because its design turned out not to be computable on the field it was run on, and one representation class has no finite test and remains open with its three requirements itemised.

The verdicts share one blocking coordinate. The pair constraint attached to D is *bilinear* — $m'u - mu' = N(m'-m)$ — where every route's decisive lemma consumes a *linear* one as a hypothesis; the h-average of the shifted-convolution world is a translation, the k-average here is a dilation, and no character family, additive or multiplicative, diagonalizes it. In the automorphic setting, shifted convolutions $\sum a(n)a(n+h)$ are controlled because the coefficients come *with* a spectral representation. μ has none, and each route below fails exactly where it tries to borrow one.

We also state one positive conjecture that any future construction must reproduce: every Möbius family probed here factorizes as a deterministic local mask, computable exactly by finite modular enumeration, times a fluctuation with no mean field and no class structure on the surviving support (Conjecture 3). Its evidence is graded: one of four supporting measurements has been independently re-implemented and reproduced; the others are cited as motivation and not as verified measurement.

This is a negative report. Net progress toward Goldbach: zero. Its purpose is to spend a search once, so that the same seventeen directions need not be re-explored, and to make explicit what a successful technique would have to supply.

1 The field, and the consumable

Let N be a large even integer. The Huang–Li reduction [1], once its demand side is evaluated [5], needs $C(N) = \sum_{n<N} \Lambda(n)\mu(N-n) = o(N)$; the natural attempt to supply it by dispersion produces the dilate field. The distilled consumable is

$$\textbf{E1 (weak form)}: \quad D(k) = \sum_{\sqrt{N}<m\le N/k} \mu(m)\mu(N-mk), \tag{1}$$

$$\sum_{k\sim K} |D(k)|^2 \;\ll\; (\log N)^{-2A-2} \sum_{k\sim K} M_k^2,$$

for $K \le N^{1/3}$ dyadic, where M_k is the length of the m-range. A fixed log-power saving is the currency; $o(1)$ and $\log\log$ savings are not consumable.

Note 1 (the normalisation is load-bearing). The normalisation in (1) is M_k^2, the *trivial* bound, not M_k, the square-root scale. The wall is $T_{II} = \sum_{k\sim K} b_k D(k)$ with $|b_k| \ll \log N$, needed at $N(\log N)^{-A}$, and Cauchy–Schwarz in k gives $\sum_k |D(k)|^2 \ll N^2/(K(\log N)^{2A+2})$ with $\sum_{k\sim K} M_k^2 \asymp N^2/K$. The distinction changes every budget by a power of N: at the square-root normalisation the demand at $N = 10^8$, $A = 1$ would be $\sum |D|^2/\sum M_k \ll 8.8 \cdot 10^{-6}$, whereas the measured ratio is of order the density of the surviving support, four orders of magnitude larger. A target must be checked against one's own measurement of the same quantity before it is used to adjudicate anything.

Measurement 2 (the surviving support, and its density)**.** The three bands are $K < k \le 2K$ for $K = 58, 116, 232$, i.e. $[59, 116]$, $[117, 232]$, $[233, 464]$ with $464 = \lfloor N^{1/3} \rfloor$ at $N = 10^8$. The convention has to be stated: the other reading of "dyadic", $K \le k < 2K$, holds the same 58, 116, 232 values of k and so is not excluded by the counts, but it gives 0.3345, 0.3278, 0.3310 for the support density below instead of 0.3303, 0.3298, 0.3320 — a shift larger than the effect the paragraph is about.

It is tempting to read that density as $\prod_p(1 - 2/p^2) = 0.3226341$, the value obtained by assuming $p^2 \mid m$ and $p^2 \mid N - mk$ cut out two distinct classes mod p^2. At $N = 10^8 = 2^8 \cdot 5^8$ that assumption fails at $p = 2$ and $p = 5$, where $p^2 \mid N$ collapses the second condition onto the first, and the true density depends on $v_p(k)$: measured exactly over all k in the three bands it ranges from 0 to 0.594330, and lands within 0.01 of 0.32263 for only 1, 1 and 3 of the 58, 116 and 232 values of k. In particular $D(k)$ vanishes identically whenever $4 \mid k$ or $25 \mid k$ — 2799 of the 9999 values $k < \sqrt{N}$, or 28.0%, and that set is exactly the vanishing set, with no other k in the bands giving $D(k) = 0$. Aggregated, the coincidence is close but not exact: $\sum_k \mathrm{supp}(k)/\sum_k M_k$ is 0.3303, 0.3298, 0.3320 on the three bands, high by 2.2 to 2.9% against the closed form, which is therefore not the right one (`audit_support_density.py`).

What the measurements establish about (1) is the weaker of the two possible readings, and it is the one the section needs: the ratio $\sum_k |D(k)|^2/\sum_k M_k$ is of order the support density and not of order $(\log N)^{-2A-2} \approx 8.8 \cdot 10^{-6}$, so the square-root normalisation is refuted by four orders of magnitude. At the correct normalisation the target is cleared with margin $(N/K)(\log N)^{-2A-2}$, which is asymptotic and is below 1 everywhere accessible; no computation is offered as evidence for it.

2 The factorization law

Conjecture 3 (the factorization law)**.** *Each Möbius family of the following kinds — prime-indexed dilate pairs $C_{k,k'} = \sum_{p\sim P} \mu(N - pk)\mu(N - pk')$, integer-indexed dilates and their pairs, and Möbius sums over thin progressions — factorizes as*

$$\textit{field} \;=\; \mathbf{M} \times \mathbf{G},$$

where $\mathbf{M}$ *is the deterministic local mask, computed exactly by finite modular enumeration from the v_q-data of (N, k, k'), and* $\mathbf{G}$ *is, on the surviving support, a fluctuation with no mean field and no class structure.*

Note 4 (the grade of the evidence). The evidence is of two kinds and they are kept apart. The E1 arm (Measurement 2) has been reproduced independently, and so has the blind mask prediction: it asserts a set equality — the field's support is empty on exactly the pairs the mask names — so a second implementation either reproduces it exactly or does not, and it does. The exact (v_2, v_3) cell predictions were attempted next and the attempt did not reach them. The remaining two supporting measurements, pair statistics and the Wishart pair spectrum, are single-witness: they are the reason the conjecture is stated, not measurements this report vouches for.

3 The negative map

Every entry below was pre-registered: the decision rule was fixed in writing before the computation ran, including the rule that would refute the hypothesis under test. Route adjudications were done in fresh context against the source papers' verbatim lemma hypotheses. The verdicts are stated without their supporting statistics; each carries its own null and its own error bar in the repository.

3.1 Adjudication of existing machinery (5)

Route	Verdict	Blocking coordinate
[4] / [2], shift $\to$ dilate	Blocked	the orthogonality factorization needs a *linear* pair constraint ($h = m - n$); the dilate constraint $m'u - mu' = N(m' - m)$ is bilinear. The h-average is a translation; the k-average is a dilation, with no diagonalizing character family.
[8] entropy decrement, k-averaged	Blocked $\times 3$	(i) no k-analogue of approximate affine invariance; (ii) the sampling prime enters the phase multiplicatively, so the sparsification fails and the surviving input is the target itself; (iii) the saving is triple-log, below spec.
[3] rerun in dilate coordinates	Blocked	the congruence coupling is a genuine isomorphism but powers only the typical-set restriction; the decoupling blocks at the same bilinear coordinate, and the uniformity slot would need an EH_μ-grade input, which is circular.
Dirichlet-polynomial fourth moment / Perron	Blocked	every log-saving mean-value pillar assumes coefficient multiplicativity in its own variable; $\mu(N - u)$ has none. Unfolding by Perron costs $T \gtrsim N^{1-o(1)}$; opening the fourth moment reproduces the binary correlation.
Partial slices	Partial	the type-I slice is already consumed; the N-*averaged* theorem is provable but lands in exceptional-set territory, which [1] cannot consume; no nontrivial fixed-N slice exists.

The common obstruction. $\mu(m)\mu(N - mk)$ couples its variables simultaneously through the product mk and the difference $N - mk$; the pair constraint is bilinear and is diagonalized by no single character family, additive or multiplicative, while each route's decisive lemma consumes exactly that diagonalization as a hypothesis.

3.2 Technique designs, kill-tested (9)

#	Design	Result (pre-registered rule applied)
K1	multiplicative Fejér kernel on the exact dilation ladder	**open**. On the type-II field the design is not computable: the $\sqrt{N}$ cut empties all but thirteen of the sixty-three orbit columns, so what is measurable there is a fifth of the orbit the design names. On the untruncated field, where the whole orbit is live, the statistic lands in the band the pre-registration reserved for "repeat at a second N before deciding".
K2	manufactured pair congruence (determinant / Kloosterman)	**dead**: congruent pairs are statistically h-blind, and the verdict is quantitative — the design would need several orders of magnitude more than its own detection floor.
K3	Wishart / operator moment method	**dead**: the second, third and fourth traces sit on the Wishart null. No sub-Wishart surplus to fund the exchange.
K4	N-average descent	**dead**: the dual field is δ-blind and m-uncorrelated.
R1	zero-spectrum visibility (explicit formula)	**dead** on the measurement, which sits below its own null. No exclusion interval is claimed, the null's spread being estimated from too few draws to support one.
R2	determinant / Kloosterman phase	**dead** on the measurement, which sits below its control. No exclusion interval is claimed; the coherent-gain arm is not blind but sits about 2.8 standard errors above its own null.
R3	character transform of the k-average	**dead by analysis**: for $K \leq N^{1/3}$ the transform deposits every component into thin progressions (moduli $\geq N^{2/3}$); Parseval forbids a statistical gain.
R4	divisor switch	**dead**: Section 4.
R5	circle method applied directly to $C(N)$	**dead, zero margin**: both pairings lie at or above the trivial bound, and the cap on the pointwise route is Parseval [6, Prop. E].

Note 5 (what a null verdict costs). A kill-test that does not fire is evidence of absence only against a stated floor, and a threshold chosen as an effect size is not a threshold. Every DEAD verdict above rests on the measurement — the statistic never sits systematically above its null — and not on a count of flagged levels; where the null's own precision could not be established, the verdict is retained but no exclusion interval is quoted. K1 is listed as open rather than closed because its design was not computable on the field it was run on, which is a defect of the run and not a result.

3.3 Representation classes (3 closed, 1 open)

Class	Test	Result
C-I abelian	rational-peak energy vs mask-null	**closed**: the abelian spectrum is mask-exact.
C-II inverse domain	special-frequency excess in the modular-inverse re-indexing	**closed on verification**: fired under eight-draw nulls, collapsed under sixty-four-draw nulls, with the permutation null concurring and the second-N replication empty.

Class	Test	Result
C-IV manufactured modularity	approximate Fricke law for $\Phi(z) = \sum t_m e(mz)$	**closed**. A true modular form drives the defect to ≈ 0, so absence is meaningful here.
C-III Motohashi type	(no finite test)	**open**, needing exactly the three items of Section 6.

4 R4 in detail: the divisor switch does not localize

Proposition 6 (perfect cancellation, untruncated)**.** *Over its full ranges the divisor switch gives the exact identity*

$$\sum_{k\geq 1} \sum_{m:\, mk\leq N-1} \mu(m)\mu(N-mk) \;=\; \sum_{u<N} \mu(N-u) \sum_{m|u} \mu(m) \;=\; \mu(N-1).$$

Proof. Exchange the order of summation and put $u = mk$; the inner sum $\sum_{m|u} \mu(m)$ is $\mathbf{1}_{u=1}$. $\square$

This is perfect cancellation: $O(1)$ for a double sum of $\sim N \log N$ terms, far beyond square root, and it is the strongest cancellation found anywhere in this programme. It is also useless, which is the content of the R4 verdict. E1 imposes two restrictions that make the inner divisor sum *incomplete* — the type-II cut $m > \sqrt{N}$ and the dyadic band $k \sim K$ — and the kill-test asked whether any of the cancellation survives them, by measuring block sums $S_B(j) = \sum_{k\in\text{block}} D(k)$ against the B-independence that Conjecture 3 predicts. It does not survive. The switch's cancellation is a property of *complete* divisor sums, and the type-II cut makes every divisor sum incomplete.

5 What the closures constrain

Any transform or invariance T that linearizes the pair constraint at a cost of at most a log power must satisfy all of the following.

- **(K2)** T cannot manufacture its congruence externally: collapse structure pays only when it arises inside an intrinsic average that is already present.
- **(K3)** T cannot bootstrap from the field's own moments: every moment consumes higher μ-correlations and the field carries no sub-Wishart surplus.
- **(K4)** T cannot descend from the N-average: its linearization is load-bearing and the k-average holds no shadow of it.
- **(R4)** T cannot be the divisor switch or a relative: the switch's cancellation is a property of complete divisor sums, and the type-II cut makes every divisor sum incomplete (Section 4).

The constraint that K1 was to have supplied — that T cannot act through divisibility subsums — is *not* asserted, K1 being open. Together with the second-round findings, characters merely relocating the difficulty into thin progressions and determinant phases not reaching their bar, *the dilate field $D(k)$* offers no coupling surface in any direction that has an existing mathematical name. The technique to be created is a spectral, or equivalent structural, representation of μ-pairs themselves, not a projection onto existing spectra — and its construction is a purely theoretical act, beyond what kill-testing can reach.

The scope of that sentence is the field of (1), in which *both* factors are Möbius and the pair constraint is bilinear. It is not a claim about every sum this programme meets. Sums of the shape $\sum_d \mu(d) \sum_m \mu^2(m)\Lambda(N-dm)w(m)$, with a single Möbius on a short variable and Λ

on the long one, are Type II bilinear forms of the classical kind and do have a literature; the difference is exactly the one Section 4 turns on, and it is why the demand side and the supply side of this problem are not interchangeable.

6 C-III: what it needs

The one representation class with no finite test is the Motohashi type. It needs exactly three items, and they are of decreasing tractability.

(1) A transform carrying the dilate average to a shift average. Since μ lives on $\mathbb{Z}$ and any such map must send finite points to finite points, the available changes of variable are exactly the integral affine ones, and no integral affine map carries one family to the other. This settles by proof what a direct probe had only measured, the off-diagonal computed directly and via the obvious re-indexing agreeing exactly, i.e. the re-indexing being circular. What remains is the summation-formula class, blocked in general by the roughness of the outer weight μ, with every named candidate in it already closed.

(2) A classification covering the type-II region. The natural bookkeeping is a Heath–Brown-type identity of level J with cut $z = M^{1/J}$, and it needs the Möbius-side variable a to satisfy $a \le M^{o(1)}$. It does not: the identity requires $z^J \ge x$, and its j-th term has $a \le z^j$, which at $j = J$ is x for every admissible (z, J) — while the weight concentrates in exactly those high-j terms. Taking the absolute weight of the j-th term to be its L^1 mass, $W_j = \binom{J}{j}\#\{(m_1..m_j, n_1..n_j) : m_i \le z \text{ squarefree}, \prod m_i \prod n_i \le x\}$, and the share to be $W_j/\sum_i W_i$, the $j \in \{6, 7, 8\}$ share at $J = 8$ reads 0.772590, 0.840039, 0.886081 at $M = 10^4$, 10^5, 10^6: the majority sits outside any region $a \le M^\eta$ with η small, and the fraction increases with M. At $M = 10^6$ the top-j share is 0.848668 at $J = 3$. The obstruction is parameter-independent, which is what carries the paragraph; the shares are quoted to indicate size, and only the rounding $z = \lceil M^{1/J} \rceil$ is admissible at all three M, since rounding down violates $z^J \ge x$ at $J = 3$ and $J = 8$ alike (`audit_hb_weight.py`).

A repair would have to bound $\sum_a \sum_b \alpha(a)\beta(b)\mu(N - abk)$ with α rough and a long, where every individual piece is trivial and all the content is cancellation across a — a type-II estimate for $\mu(N - \cdot)$, i.e. the wall itself. Every identity decomposing μ produces such a term; one whose every piece had either a long free variable or a divisor-structured rough coefficient would dispose of the parity obstruction. **Completing the construction and breaking the wall are the same task.**

(3) Quantitative averaged Chowla. Shift-averaged binary $\mu\mu$ correlations at a *fixed* log-power saving, against a best known of $(\log)^{1-c}$. This is a named external open problem and is where C-III stands.

That third item sets the character of the obstruction. The adjudication's central finding is that the **dilate** average admits no diagonalizing character family; but the **shift** average is precisely the home ground of the Matomäki–Radziwiłł–Tao machinery. If a legitimate transform converting one into the other exists, what remains is quantitative strength inside an active area rather than the absence of any coupling surface.

7 Methodology

Four rules were followed throughout, and they are the reason the negative results are believable.

- **Pre-registration.** Every design's decision rule was written before its computation ran, including the rule that would refute the hypothesis under test.

- **Adversarial review in fresh context**, against the source papers' verbatim lemma hypotheses rather than against summaries of them, and repeated. Coverage is uneven and is stated rather than rounded up: material carried forward from the first version of this programme has been through three independent reviews; statements first written later have had fewer. The measured detection rate of a single review of this kind is about one half, so nothing here should be read as certified.

- **Nulls before thresholds.** A threshold means nothing until the spread of the quantity it judges has been measured, and a null must preserve the structure of the field it is a null for. A null built by shuffling a correlated field is not a null for a statistic that reads its correlation.

- **Weights and fields before comparisons.** Two summaries of one object are not comparable until each one's weight is stated (Note 1), and a measured figure is not interpretable until the range it was measured on is stated.

Three traps are worth recording because they are traps and not slips.

- *The $(q, N) > 1$ main-term trap.* Assigning a main term $T_m/\varphi(q)$ to a residue class with $(q, N) > 1$ — which is degenerate, its true contribution being $O(\log^2 N)$ — shifts the density of a whole computation by exactly $N/\varphi(N)$. Carried into the $\log k$ branch of the demand side, that error produces an apparent *refutation* of EH_μ. It is the most plausible false positive in this circle of ideas.

- *Nulls estimated from too few draws* inflate maxima across a family of tests. One representation-class hit survived eight-draw nulls and died under sixty-four-draw nulls plus replication.

- *Local factors evaluated at the wrong modulus.* Dividing by $W(N)$ where the definition calls for $V(N)$ changes the answer by exactly $\mathfrak{A}(N)$ [7].

8 Summary

- Seventeen routes into the dilate field were pre-registered and run: five adjudications, nine kill-tests, four representation classes. Fifteen close, K1 is reopened, and C-III is open with its three requirements itemised.

- The common blocking coordinate is bilinearity of the pair constraint. Every route's decisive lemma consumes a linear constraint as a hypothesis.

- Proposition 6: the untruncated dilate double sum cancels perfectly, to $\mu(N-1)$. The cancellation is a property of complete divisor sums and does not survive the type-II cut.

- Conjecture 3 is what any future construction must reproduce: a deterministic modular mask times a structureless fluctuation. Its supporting measurements are graded in Note 4.

- Net progress toward Goldbach: zero. The deliverable is the map, and the statement of what a successful technique must supply.

References

[1] J.-J. Huang and H. Li, *On the connection between the Goldbach conjecture and the Elliott–Halberstam conjecture*, arXiv:2005.03811 [math.NT]; v1 (May 2020), v2 (August 2022). Also published in a Springer volume, `doi:10.1007/978-3-030-67996-5_17`. References

here are to the arXiv version, which we have read, and not to the published version, which we have not.

[2] J. D. Lichtman, *Averages of the Möbius function on shifted primes*, arXiv:2009.08969 [math.NT], 2020.

[3] J. D. Lichtman (with an appendix by S. Drappeau), *Primes in arithmetic progressions to large moduli, and Goldbach beyond the square-root barrier*, arXiv:2309.08522 [math.NT], 2023.

[4] K. Matomäki, M. Radziwiłł and T. Tao, *An averaged form of Chowla's conjecture*, Algebra Number Theory **9** (2015), 2167–2196.

[5] *An unconditional bound for a Möbius-weighted correlation sum in fixed residue classes*, companion paper.

[6] *No weight extracts $\sum_{n<N} \Lambda(n)\mu(N-n)$: a no-go for the divisor-switch route*, companion paper.

[7] *The exact second moment of $\sum_{n<N} \Lambda(n)\mu(N-n)$, and why Chowla's conjecture does not control it*, companion paper.

[8] T. Tao, *The logarithmically averaged Chowla and Elliott conjectures for two-point correlations*, Forum Math. Pi **4** (2016), e8.

A Reproducibility

Script	Result file	Supports
`audit_support_density.py`	`audit_support_density.txt`	Measurement 2
`audit_hb_weight.py`	`audit_hb_weight.txt`	Section 6, item (2)
`audit_r4_blocks.py`	`audit_r4_blocks.txt`	Section 4
`audit_sieve.py`	`audit_sieve.txt`	the sieve variants used throughout

The verdicts of Section 3 are stated without their supporting statistics. Each design's decision rule, its null, and its result file are recorded in the repository accompanying this report; the table above lists only the measurements quoted in this document.